A VARIANT OF THE DERIVED PROJECTION FORMULA FOR A MORPHISM OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument. It displays the module-level identity and then states, in the author's own words, that the lemma follows from this and the definitions; that closing sentence is retained verbatim inside the counted proof span and is not replaced or expanded by this card.

Editorial note. The counted spans contain no label reference and no bibliography citation, so the card carries no navigation stub and no bibliography excerpt.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this one, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Morphism of ringed topoi and the functors f^{-1}, f^* and Lf^*. Same chapter, the inverse image f^{-1} of sheaves and the derived pullback Lf^* on modules over a ringed topos. Supplies the functors appearing on both sides of the asserted isomorphism.
- Derived tensor product of complexes of modules over a ringed topos. Same chapter, the derived tensor product tensor^L and its computation from K-flat representatives. Supplies the objects compared and the passage from the module-level identity to complexes.
- The identification f^*G = O tensor over f^{-1}O' with f^{-1}G. Same chapter, the description of the pullback of a module as a tensor product over the inverse image of the structure sheaf. Supplies the displayed identity whose proof is the whole content of the argument.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. There is a canonical bifunctorial isomorphism*

$$\mathcal{F}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} Lf^*\mathcal{G}^\bullet = \mathcal{F}^\bullet \otimes^{\mathbf{L}}_{f^{-1}\mathcal{O}'} f^{-1}\mathcal{G}^\bullet$$

for $\mathcal{F}^\bullet$ in $D(\mathcal{O})$ and $\mathcal{G}^\bullet$ in $D(\mathcal{O}')$.

Proof. Let $\mathcal{F}$ be an $\mathcal{O}$-module and let $\mathcal{G}$ be an $\mathcal{O}'$-module. Then $\mathcal{F} \otimes_{\mathcal{O}} f^*\mathcal{G} = \mathcal{F} \otimes_{f^{-1}\mathcal{O}'} f^{-1}\mathcal{G}$ because $f^*\mathcal{G} = \mathcal{O} \otimes_{f^{-1}\mathcal{O}'} f^{-1}\mathcal{G}$. The lemma follows from this and the definitions. □

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